UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWORKS

SUBMITTED BY & TO

NAME : RIFAQ AJMAL

REG NO : 23MDBCS 435

SECTION : A

SEMESTER : 4th

SUBMITTED T0 : NAJEEB SIR

LAB # 4

Lab 4 – Capture and Analyze DNS Traffic Using Wireshark

Introduction

In this lab, you will capture some DNS traffic and analyze it using Wireshark.

Lab Tasks

- Launch Wireshark and start capturing traffic.
- Open a new website. This should create some DNS traffic.
- Stop the capture.
- Filter the captured traffic based on DNS.
- Identify various types of DNS records in the DNS messages and paste snapshot below.
- Label each type such as Type A, Type CNAME etc.

- **DNS RECORDS QUERY AND RESPONSES:**

1035	1.443024	192.168.187.252	192.168.187.35	DNS	95	Standard query response 0x4d4b A clients4.google.com A 142.250.201.142
1279	2.242302	192.168.187.35	192.168.187.252	DNS	94	Standard query 0x8032 A kv601.prod.do.dsp.mp.microsoft.com
1324	2.286752	192.168.187.252	192.168.187.35	DNS	204	Standard query response 0x8032 A kv601.prod.do.dsp.mp.microsoft.com CNAME kv601.prod.do.dsp…
2311	3.998592	192.168.187.35	192.168.187.252	DNS	75	Standard query 0xde3f A www.youtube.com
2312	3.998828	192.168.187.35	192.168.187.252	DNS	75	Standard query 0x1c0d HTTPS www.youtube.com
2313	4.040258	192.168.187.252	192.168.187.35	DNS	171	Standard query response 0xde3f A www.youtube.com A 192.178.24.238 A 192.178.24.206 A 192.17…
2314	4.042238	192.168.187.252	192.168.187.35	DNS	124	Standard query response 0x1c0d HTTPS www.youtube.com CNAME youtube-ui.l.google.com HTTPS
3119	4.375627	192.168.187.35	192.168.187.252	DNS	95	Standard query 0x5283 A optimizationguide-pa.googleapis.com
3120	4.375788	192.168.187.35	192.168.187.252	DNS	95	Standard query 0x95d0 HTTPS optimizationguide-pa.googleapis.com
3121	4.377036	192.168.187.35	192.168.187.252	DNS	71	Standard query 0x65a4 A youtube.com
3122	4.377183	192.168.187.35	192.168.187.252	DNS	71	Standard query 0x91c0 HTTPS youtube.com
3206	4.420246	192.168.187.252	192.168.187.35	DNS	191	Standard query response 0x5283 A optimizationguide-pa.googleapis.com A 192.178.24.234 A 172…
3207	4.420476	192.168.187.252	192.168.187.35	DNS	152	Standard query response 0x95d0 HTTPS optimizationguide-pa.googleapis.com SOA ns1.google.com
3208	4.420607	192.168.187.252	192.168.187.35	DNS	87	Standard query response 0x65a4 A youtube.com A 192.178.24.206
3209	4.420618	192.168.187.252	192.168.187.35	DNS	86	Standard query response 0x91c0 HTTPS youtube.com HTTPS
3777	5.761605	192.168.187.35	192.168.187.252	DNS	71	Standard query 0xc573 A i.ytimg.com
3778	5.762145	192.168.187.35	192.168.187.252	DNS	71	Standard query 0xab8f HTTPS i.ytimg.com
3779	5.773860	192.168.187.35	192.168.187.252	DNS	77	Standard query 0x8c7c A fonts.gstatic.com
3780	5.774335	192.168.187.35	192.168.187.252	DNS	77	Standard query 0x5d4c HTTPS fonts.gstatic.com
3781	5.811361	192.168.187.252	192.168.187.35	DNS	93	Standard query response 0x8c7c A fonts.gstatic.com A 192.178.24.227
3782	5.811465	192.168.187.252	192.168.187.35	DNS	167	Standard query response 0xc573 A i.ytimg.com A 172.217.169.246 A 192.178.24.214 A 142.250.2…
3783	5.815239	192.168.187.252	192.168.187.35	DNS	128	Standard query response 0xab8f HTTPS i.ytimg.com SOA ns1.google.com
3784	5.815364	192.168.187.252	192.168.187.35	DNS	102	Standard query response 0x5d4c HTTPS fonts.gstatic.com HTTPS
3799	6.055725	192.168.187.35	192.168.187.252	DNS	80	Standard query 0x6010 A fonts.googleapis.com
3800	6.055890	192.168.187.35	192.168.187.252	DNS	80	Standard query 0xf65d HTTPS fonts.googleapis.com
3810	6.099489	192.168.187.252	192.168.187.35	DNS	137	Standard query response 0xf65d HTTPS fonts.googleapis.com SOA ns1.google.com

- **LABELED**

*Ethernet 8

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
28	0.790048	192.168.187.35	192.168.187.252	DNS	79	Standard query 0x4d4b A clients4.google.com
966	1.418790	192.168.187.252	192.168.187.35	DNS	95	Standard query response 0x3290 A clients4.google.com A 192.178.24.206
967	1.418884	192.168.187.252	192.168.187.35	DNS	153	Standard query response 0xfdb2 HTTPS clients4.google.com CNAME clients.l.google.com SOA ns1…
983	1.424687	192.168.187.252	192.168.187.35	DNS	95	Standard query response 0x5173 A clients4.google.com A 142.250.201.142
986	1.424906	192.168.187.252	192.168.187.35	DNS	153	Standard query response 0x4ee6 HTTPS clients4.google.com CNAME clients.l.google.com SOA ns1…
1035	1.443024	192.168.187.252	192.168.187.35	DNS	95	Standard query response 0x4d4b A clients4.google.com A 142.250.201.142
1279	2.242302	192.168.187.35	192.168.187.252	DNS	94	Standard query 0x8032 A kv601.prod.do.dsp.mp.microsoft.com
1324	2.286752	192.168.187.252	192.168.187.35	DNS	204	Standard query response 0x8032 A kv601.prod.do.dsp.mp.microsoft.com CNAME kv601.prod.do.dsp…
2311	3.998592	192.168.187.35	192.168.187.252	DNS	75	Standard query 0xde3f A www.youtube.com
2312	3.998828	192.168.187.35	192.168.187.252	DNS	75	Standard query 0x1c0d HTTPS www.youtube.com
2313	4.040258	192.168.187.252	192.168.187.35	DNS	171	Standard query response 0xde3f A www.youtube.com A 192.178.24.238 A 192.178.24.206 A 192.17…
2314	4.042238	192.168.187.252	192.168.187.35	DNS	124	Standard query response 0x1c0d HTTPS www.youtube.com CNAME youtube-ui.l.google.com HTTPS
3119	4.375627	192.168.187.35	192.168.187.252	DNS	95	Standard query 0x5283 A optimizationguide-pa.googleapis.com
3120	4.375788	192.168.187.35	192.168.187.252	DNS	95	Standard query 0x95d0 HTTPS optimizationguide-pa.googleapis.com
3121	4.377036	192.168.187.35	192.168.187.252	DNS	71	Standard query 0x65a4 A youtube.com
3122	4.377183	192.168.187.35	192.168.187.252	DNS	71	Standard query 0x91c0 HTTPS youtube.com

C-NAME

A